

BIOL 220 Probability

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Learning Objectives

1. Define random trial, event, and outcome
2. Define probability
3. Describe discrete and continuous probability distributions

Next lecture:

3. More on discrete and continuous probability distributions
4. Determine if events are mutually exclusive
5. Determine if events are independent

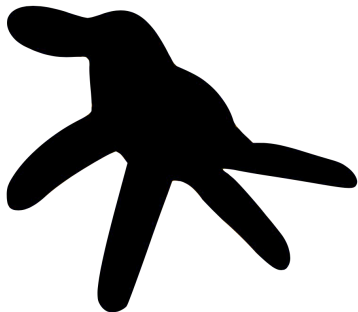
Define random trial

- What's going to happen tomorrow?
- If I flip a coin—heads or tails?
- If I roll a die, what number comes up?
- If I measure a fish, how long is it?
- If I conduct an insect survey in the Ko olau Mountains, how many species will I find?
- What is the effect of a new drug on a patient enrolled in a clinical trial?

These are all random trials

Define **random trial**

*A **random trial** is a process or experiment that has two or more possible outcomes whose occurrence cannot be predicted with certainty*



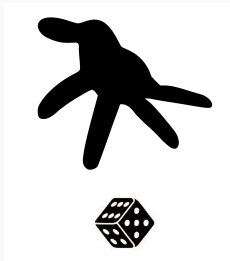
Define **event** and **outcome**

*An **event** is one observation of a random trial*

*The **outcomes** of a random trial describe all the possible events*

Define **random trial**, **event**, and **outcome**

The **random trial**:
you roll the die



An **event**:
you get a 5



All possible **outcomes**:

all integers 1—6

Define probability

*The probability of an event is the proportion of times that event would occur if we repeated a random trial over and over again under the same conditions. Probability **always** ranges between 0 and 1*

Define **probability**

With a fair die, every outcome has equal probability:



$$Pr(\text{rolling a } 1) = 1/6$$



$$Pr(\text{rolling a } 2) = 1/6$$



$$Pr(\text{rolling a } 3) = 1/6$$



$$Pr(\text{rolling a } 4) = 1/6$$



$$Pr(\text{rolling a } 5) = 1/6$$



$$Pr(\text{rolling a } 6) = 1/6$$

Define **probability**

Probability is just counting, sometimes *very complex* counting.

But with a die it is simple counting:

$$Pr(\text{rolling a } 5) = 1/6$$

because there is only **1** way to roll a 5, and there are **6** possible outcomes

Probability is all the ways you could get a specific event, divided by the number of possible outcomes

Define **probability**

Probability is all the ways you could get a specific event, divided by the number of possible outcomes

Consider rolling **2 dice** what are all the possible outcomes?

Pause & Think

Why is 7 a lucky number in many cultures?

Define **probability**

Probability is all the ways you could get a specific event, divided by the number of possible outcomes

all the ways to get 7 from 2 dice:

(1, 6); (6, 1); (2, 5); (5, 2)

(3, 4); (4, 3)

6

number of possible outcomes of 2 dice:

6 * 6

probability of rolling a 7:

6 / 36 # = 1/6

Define **probability**

Probability is all the ways you could get a specific event, divided by the number of possible outcomes

```
# simulate it
nroll <- 10000
crap <- replicate(nroll, {
  d1 <- sample(6, 1)
  d2 <- sample(6, 1)

  d1 + d2
})

sum(crap == 7) / nroll # this should be about 1/6
```

Define **probability**

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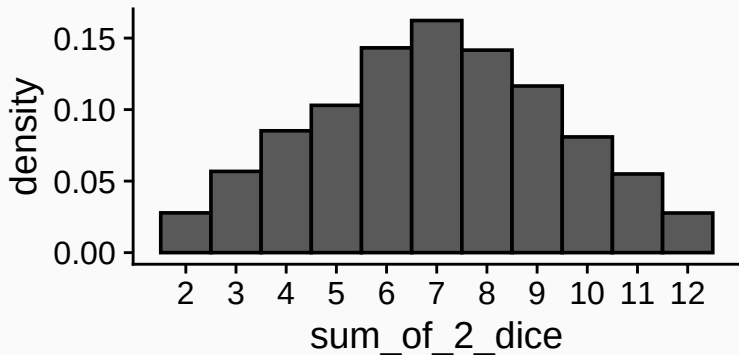
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})
```

Instead of looking at one probability:

```
sum(crap == 7) / nroll # this should be about 1/6
```

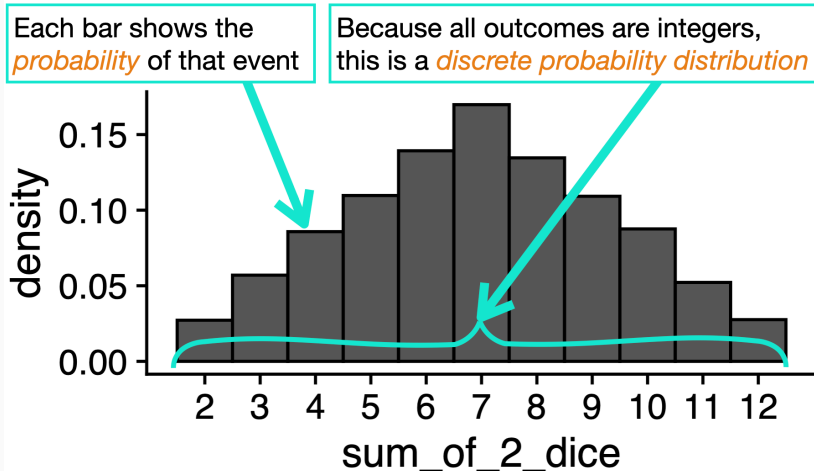
Define **probability distribution**

We could make a histogram-looking thing of all probabilities:



This histogram-looking thing is called a **probability distribution**

Define **probability distribution**



This histogram-looking thing is called a **discrete probability distribution**

Why do we care?

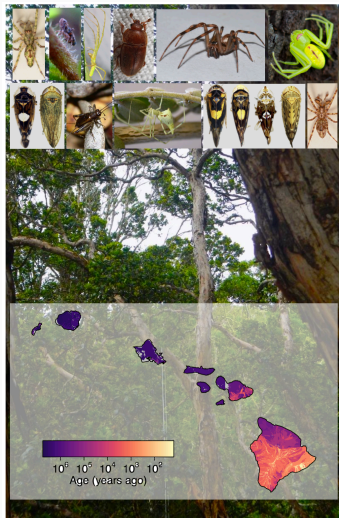
Probability is everywhere!

<https://www.youtube.com/watch?v=zJN5W-EreVs&ab>

Why do we care?

Probability is everywhere!

*If I conduct an insect survey,
how many species will I find?*



Why do we care?

Probability is everywhere!

If I conduct an insect survey, how many species will I find? What are all the outcomes:

Process	Event
No populations establish	$S = 0$
A population establishes and goes extinct	$S = 0$
2 population establish and 1 goes extinct	$S = 1$
...	...
N populations establish, M go extinct	$S = N - M$
...	...
N populations establish, none go extinct	$S = N$
...	...

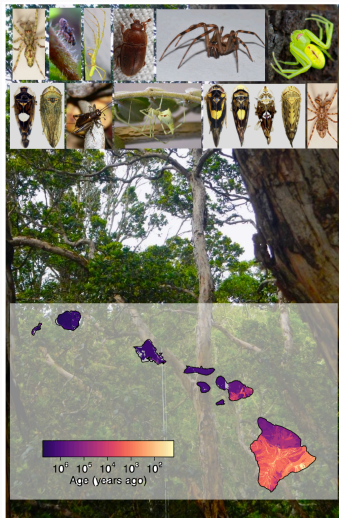
Complex counting!

Why do we care?

Probability is everywhere!

*If I conduct an insect survey,
how many species will I find?*

Different forces in nature, like
speciation or species
interactions, change the
probability of different
processes. In effect, those forces
load the dice differently

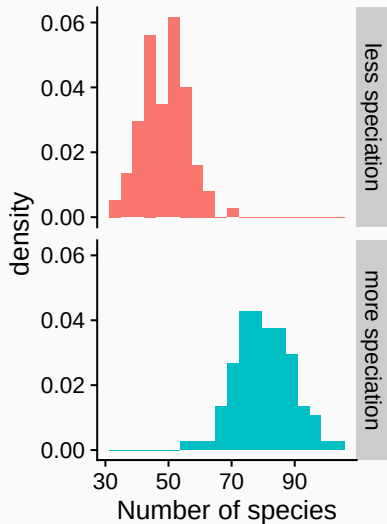


Why do we care?

Probability is everywhere!

*If I conduct an insect survey,
how many species will I find?*

Different forces in nature, like speciation or species interactions, change the probability of different processes. In effect, those forces *load the dice* differently



Why do we care?

Remember noni?

Suppose we run a clinical study of the effect of noni juice on inflammatory arthritis. We randomly assign half the patients to receive 10mg noni extract and the other half to receive a placebo. We measure change in erythrocyte sedimentation rate (*sed rate*) in all patients before and after treatment (either 10mg noni or placebo)



Why do we care?

Remember noni?

Noni is a crop brought to Hawai'i by Kānaka Moali. The 'ike of its use comes from inter-generational kilo.

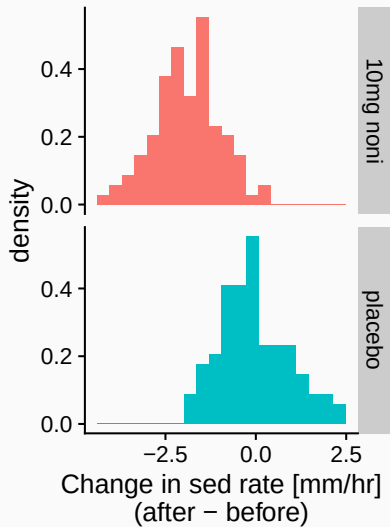
- Who gets to decide whether/how to commodify noni?
- This clinical trial should be led by Native Hawaiian practitioners in collaboration with scholars who may or may not also be Native Hawaiian



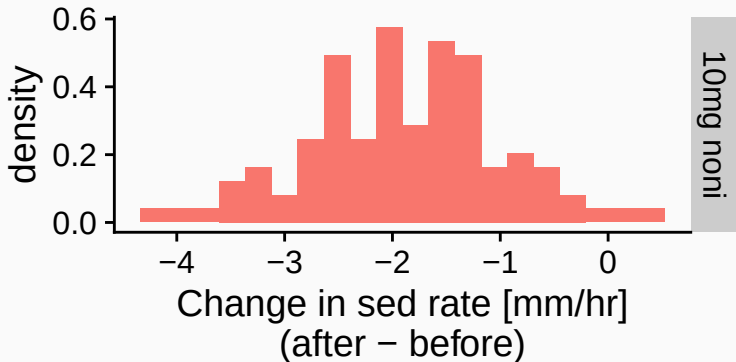
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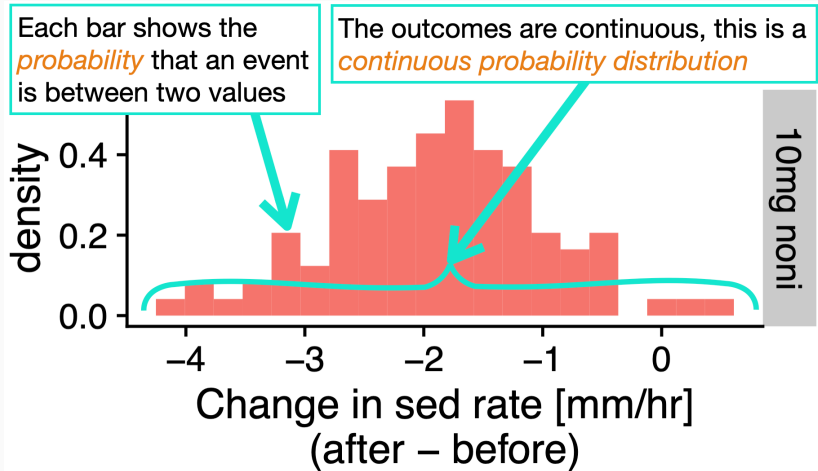
Define continuous probability distribution



*Change in sed rate [mm/hr] is **not** an integer*

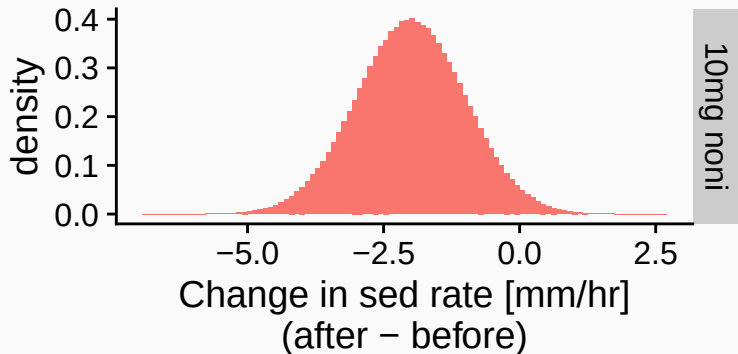
What do we do?!

Define continuous probability distribution



Define **continuous probability distribution**

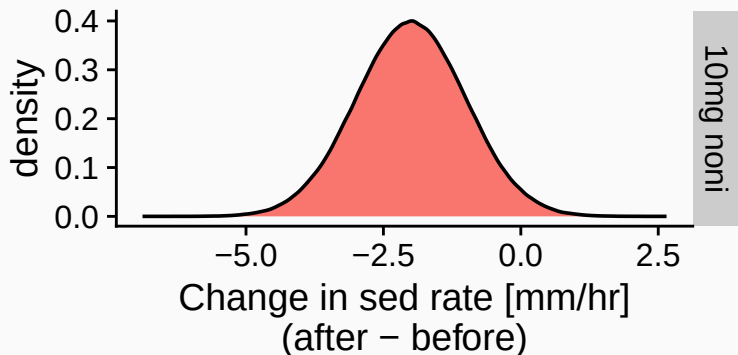
Imagine we had millions of data points, and we could draw very narrow bins



We would get a very smooth looking distribution

Define continuous probability distribution

Continuous probability distributions are typically visualized as smooth curves, *without bars*



Define continuous probability distribution

Continuous probability distributions are typically visualized as smooth curves, *without bars*

